

Climate policies exclusively centered in the Global North harm the economy of the Global South

Policy lessons of a two-region input-output ecological macroeconomic
model

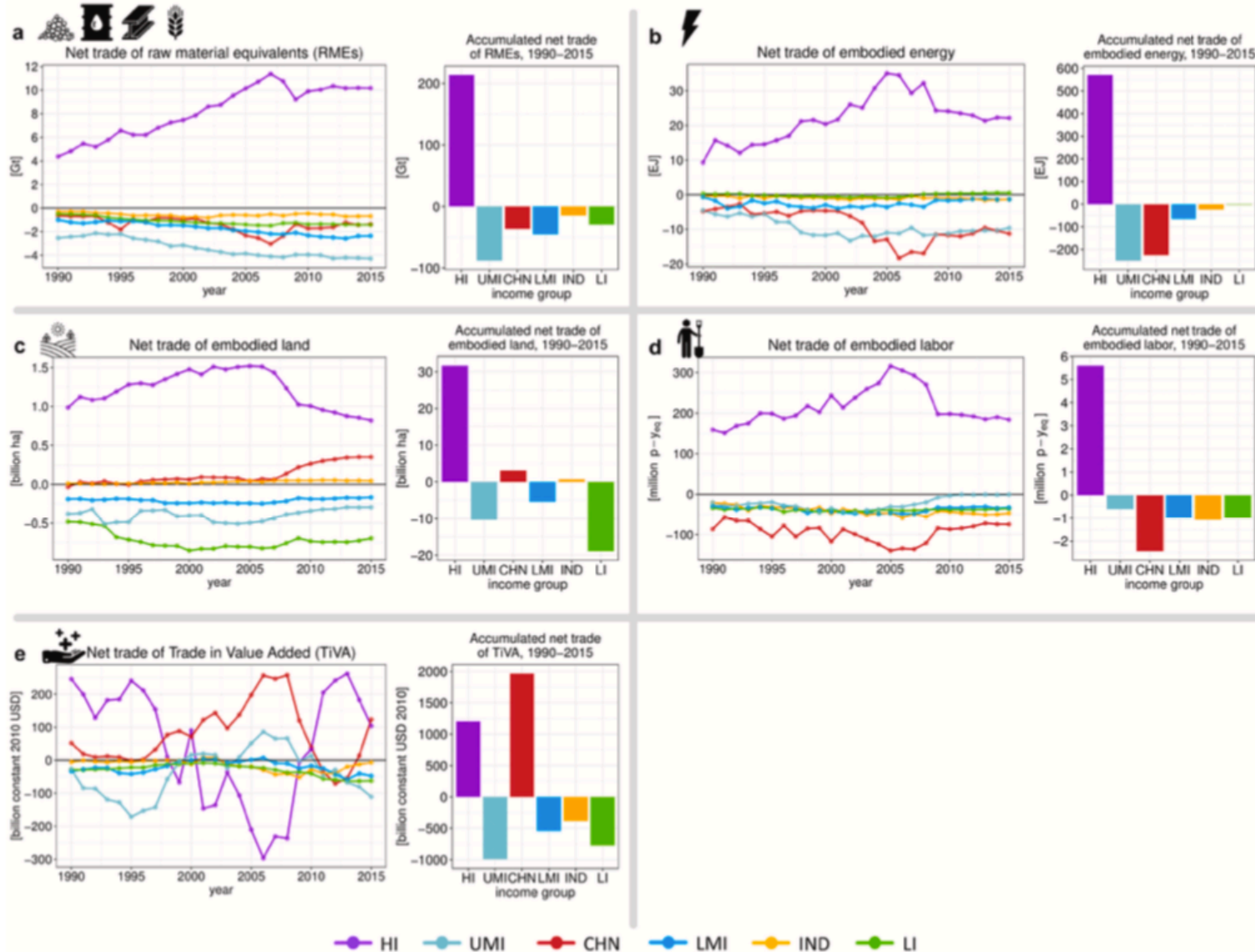
Oriol Vallès Codina

Introduction

Outline

- ECO-MRIO-SFC modeling as a tool to understand the very **complex impact of (carbon) taxes and tariffs on global value chains** from a socio-economic perspective: ecological, macroeconomic and distributional implications for class, gender, race / region
- Trade wars and global value chain fragmentation/regionalization
- **Challenge** how to overcome ecological unequal exchange?
- Current model shows ecological and economic impacts are best in cooperative models
- Implications: competition vs cooperation in trade (Blecker 2025)

Motivation: Exchange is Unequal



Starting Point: the JUST2CE project

- **JUST2CE** = A Just Transition to the Circular Economy
- EU-funded project proposing an alternative way of looking at CE
- Most projects have focused on *how* to produce circularity. JUST2CE focuses on *what* (democracy, participation, gender, global justice)
- Two milestones / WP5 deliverables:
 - **Systematic review** of macro models assessing CE transition (Feverieiro et al 2025)
 - **Formal model(s)** to simulate & compare CE policies and transition scenarios (Vallès Codina et al., 2026?)

Systematic Literature Review



Ecological Economics

Volume 236, October 2025, 108669



Macroeconomic models for assessing the transition towards a circular economy: A systematic review

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Marco Veronese Passarella ^{d e}

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<https://doi.org/10.1016/j.ecolecon.2025.108669>

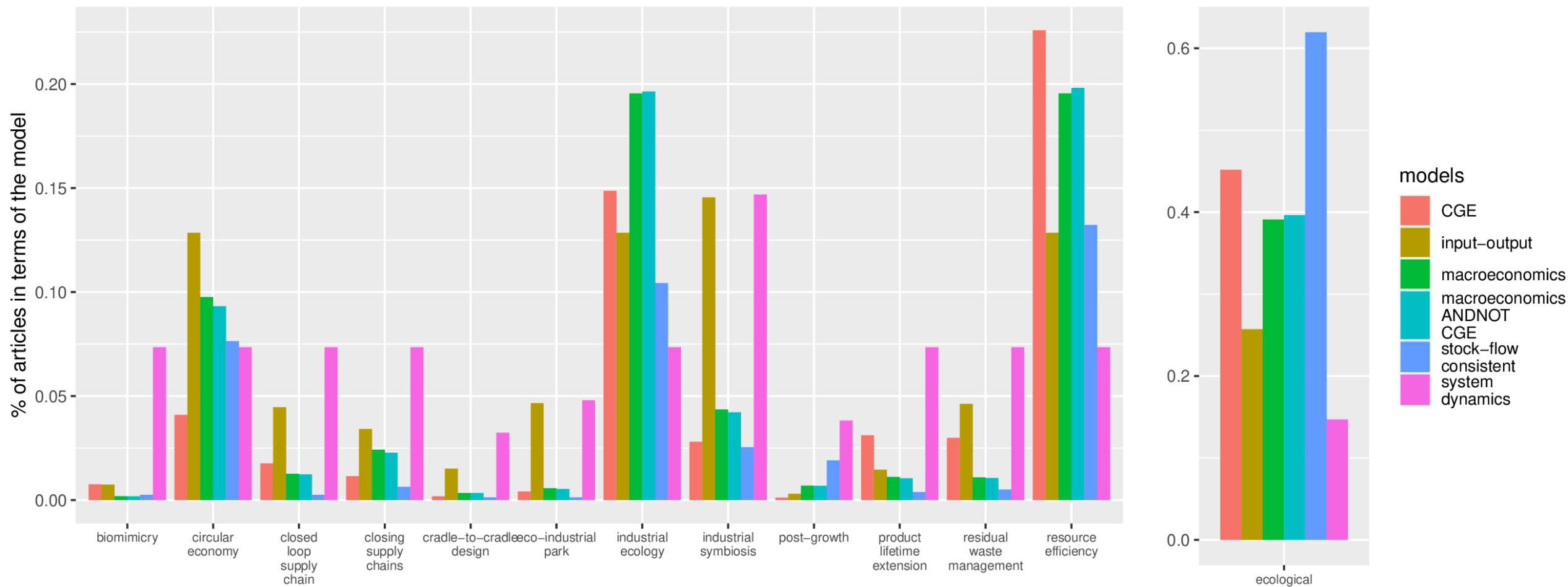
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Abstract

The Circular Economy (CE) paradigm has gained traction in both academic discourse and industrial practice. While a transition towards a CE is generally associated with more sustainable futures, less is known about its socio-economic feasibility. This article provides a systematic literature review of contributions to macroeconomic modelling which evaluate environmental and socio-economic impacts of CE interventions (classified in terms of closing supply chains, resource efficiency, residual waste management, and product lifetime extension). Differences in modelling approaches (Leontief input-output, macroeconometric input-output, and computable general

Systematic Literature Review (Simplified)

Emphasis on Circular Economy Concepts by Model



Citation Network of Ecological Macroeconomics

Main Gaps Identified

- **Modelling of Rebound Effects** — limited treatment of demand/impact changes via prices, jobs, income
- **Transitional Dynamics** — IO analysis typically static
- **Limited Socio-Economic Coverage** — often only employment considered
- **Technology Innovation & Diffusion; Demand Assumptions** — low-labor-cost tech may be preferred
- **North–South / Core–Periphery Ecological Unequal Exchange** (Dorninger 2021, Hickel)

ECO-MRIO-SFC Modeling

The Model: Basic Features

Main tools in (theory-to-data) macroeconomics:

1. **DSGE (and growth) models** — *Pros*: mainstream, simple story. *Cons*: “small”, unique optimal equilibrium
2. **CGE models (and IAMs)** — *Pros*: many variables, cross-industry links. *Cons*: unique optimal equilibrium, no finance

Cross-breeding two alternatives (*Hardt & O’Neill 2017*):

- **Leontief IO Models** — *Pros*: many variables, cross-industry links. *Cons*: static, no finance
- **SFC Models** — *Pros*: dynamics, finance. *Cons*: homogeneous output

Theoretical Antecedents

- Increasing use of IO–SFC models; yet none focuses on CE:
- *Berg, Hartley & Richters (2015)*. A stock-flow consistent input-output model with applications to energy price shocks, interest rates, and heat emissions.
- **Other contributions** *Naqvi (2015); Naqvi and Stockhammer (2018); Valdecantos & Valentini (2017); Nieto et al (2019); Jackson & Jackson (2021); Cordier et al. (2015); Jackson et al. (2014)*
- MEDEAS, Eurogreen, REAL, ToBe... increasingly very sophisticated one-region models

ECO-MRIO-SFC Model

- a. Two-country macro frame from standard SFC models (*Godley & Lavoie 2007*):
- **Six sectors:** households, production firms, government, commercial banks, central bank
 - **Four assets:** cash, bank deposits, shares, government bills (+ advances)
 - **Only loans to firms** (no personal loans)
 - **Fixed capital**, no inventories

- b. Flexible IO structure: **currently 50+ industries**
- c. Identification: calibration to target final demand components
- d. Solution: numerical simulations (R), 100 periods, 100 iterations

Final Demand and Gross Output: Constant Sector Shares

Let us consider a 3×3 production. The **final demand** vector is:

$$\mathbf{d} = \begin{bmatrix} \beta_1 \\ \beta_2 \\ \beta_3 \end{bmatrix} c + \begin{bmatrix} l_1 \\ l_2 \\ l_3 \end{bmatrix} i_d + \begin{bmatrix} \sigma_1 \\ \sigma_2 \\ \sigma_3 \end{bmatrix} gov + \begin{bmatrix} \eta_1 \\ \eta_2 \\ \eta_3 \end{bmatrix} exp$$

The **gross output** vector is:

$$\mathbf{x} = \mathbf{Ax} + \mathbf{d} = (\mathbf{I} - \mathbf{A})^{-1} \mathbf{d}, \quad \mathbf{A} = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix}$$

Classical Approach to Prices

For N regions, K industries per region, and $i = 1, \dots, KN$ industries:

$$p_{i,t} = (1 + \mu) \left[wl_i + \sum_{j=1}^{KN} p_{j,t-1} (A_{ji} + \delta_j K_{ji}) \right]$$

Prices are set sequentially as a markup over realised unit costs; under stationarity, the price vector converges to the fixed point of the corresponding IO price operator.

Carbon Taxation

Carbon taxes are internalized as a production cost for all commodities:

$$p' = p(1 + \tau)$$

where $p = w l B$ is the price before tax, p' is the price with tax, and τ is the product of emission intensity times the carbon price. In scalar notation, we write $p_i = w \sum_j l_j b_{ji}$ so that adding the carbon price yields:

$$p'_i = w(1 + \tau_i) \sum_j l_j b_{ji}$$

(Carbon) Tariffs 1

We need to exploit the $KN \times KN$ dimensionality of global matrix of input-output coefficients A and thus the augmented Leontief matrix B . For $K = N = 2$, 2 products and 2 regions, labeled 1 to 4:

$$p = w(l_1, l_2, l_3, l_4) \begin{pmatrix} b_{11} & b_{12} & b_{13} & b_{14} \\ b_{21} & b_{22} & b_{23} & b_{24} \\ b_{31} & b_{32} & b_{33} & b_{34} \\ b_{41} & b_{42} & b_{43} & b_{44} \end{pmatrix}$$

(Carbon) Tariffs 2

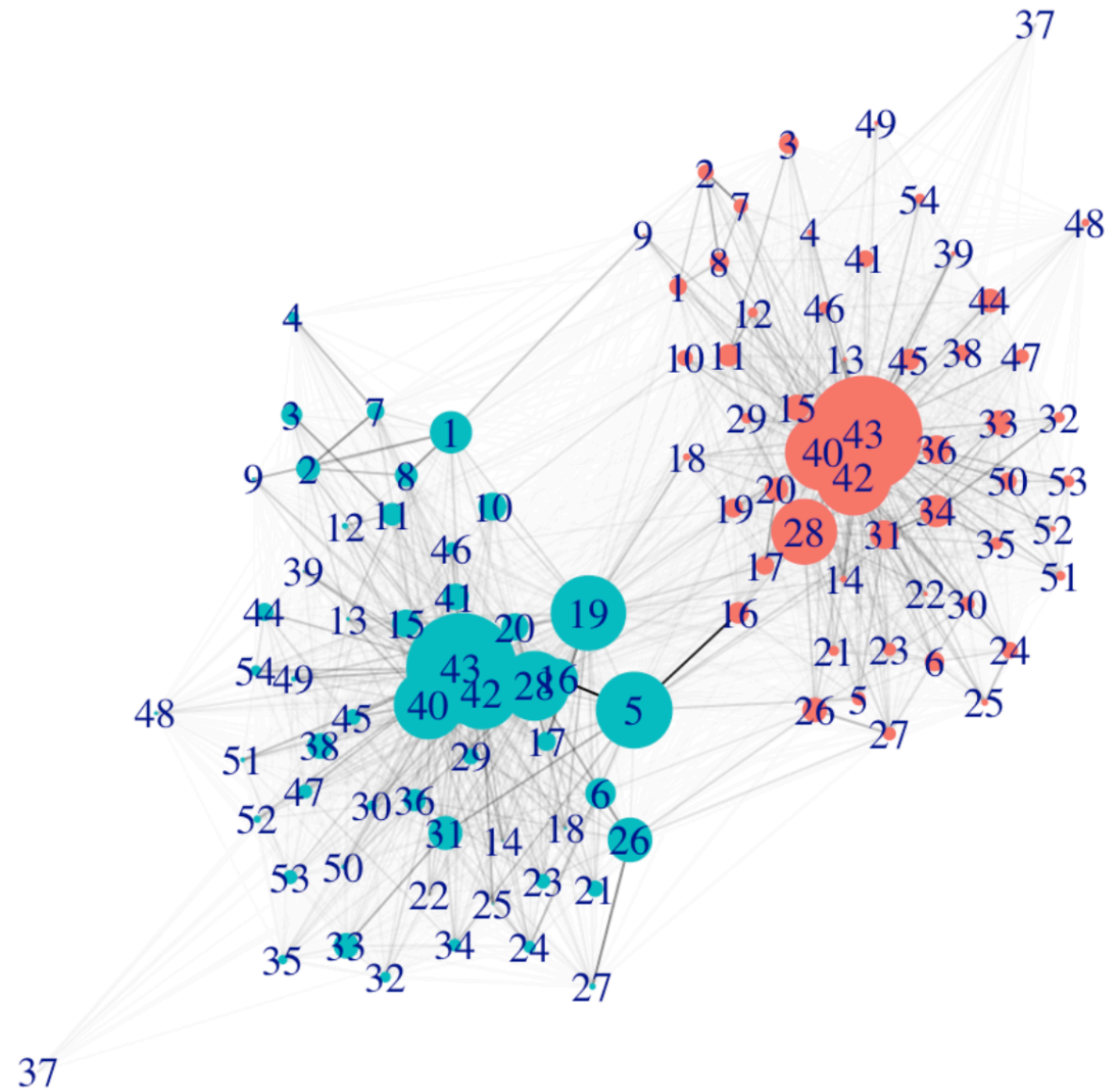
In the carbon taxation case, we defined $1 + \tau$ as a KN -vector, but it can also be thought of a diagonal matrix with zeros in the off-diagonal. For the tariffs case, operate precisely on the off-diagonal matrices that refer to interregional trade. If region 1 imposes a tariff (τ_3, τ_4) on products 1 and 2 of region 2, we multiply coefficients b_{ij} by $(1 + \tau_i)$ for $i = 3, 4$ and $j = 1, 2$:

$$p' = w(l_1, l_2, l_3, l_4) \begin{pmatrix} b_{11} & b_{12} & b_{13} & b_{14} \\ b_{21} & b_{22} & b_{23} & b_{24} \\ b_{31}(1 + \tau_3) & b_{32}(1 + \tau_3) & b_{33} & b_{34} \\ b_{41}(1 + \tau_4) & b_{42}(1 + \tau_4) & b_{43} & b_{44} \end{pmatrix}$$

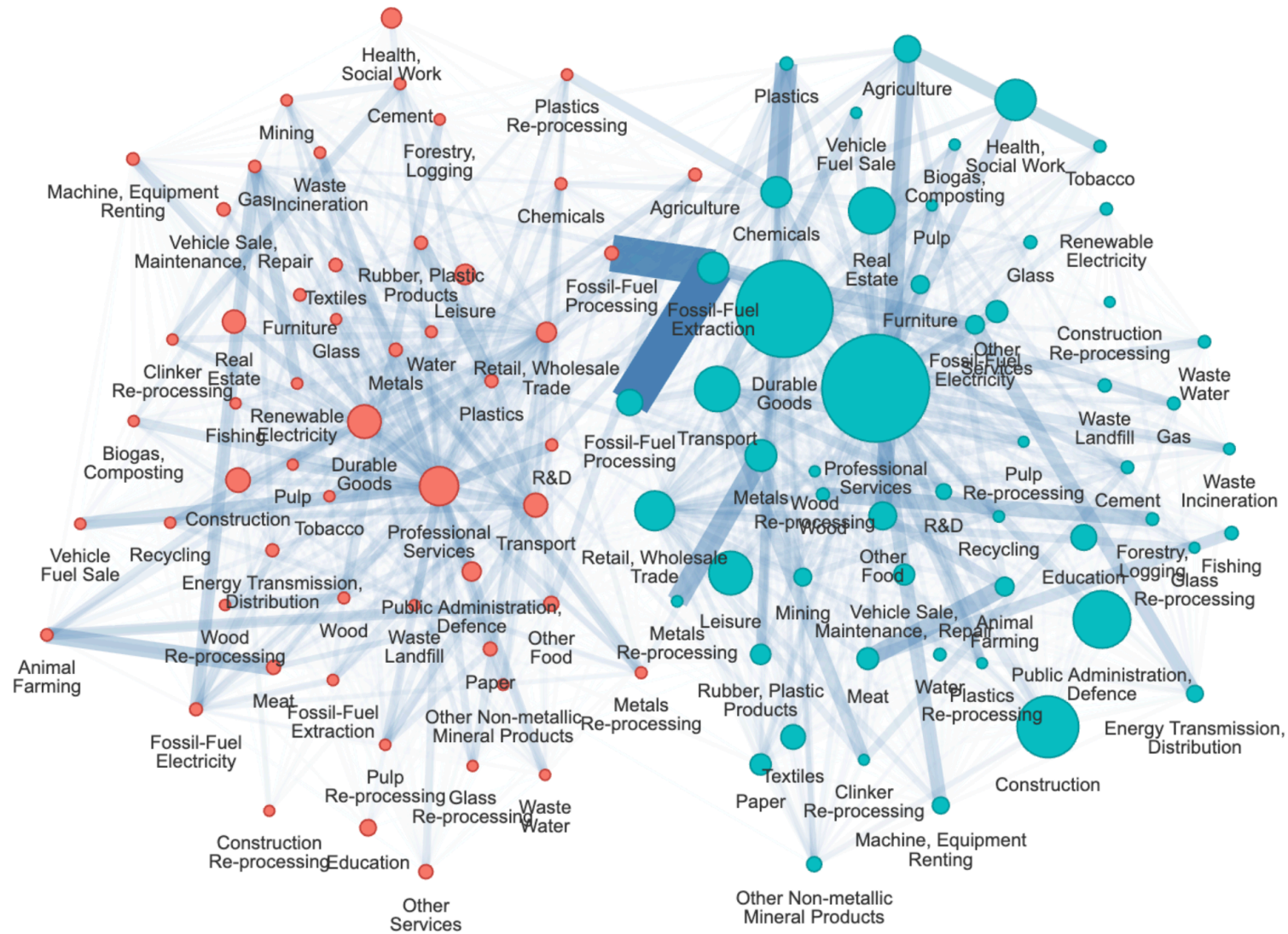
Ecological Unequal Exchange: EU (red) vs RoW (green)

Table of Industries

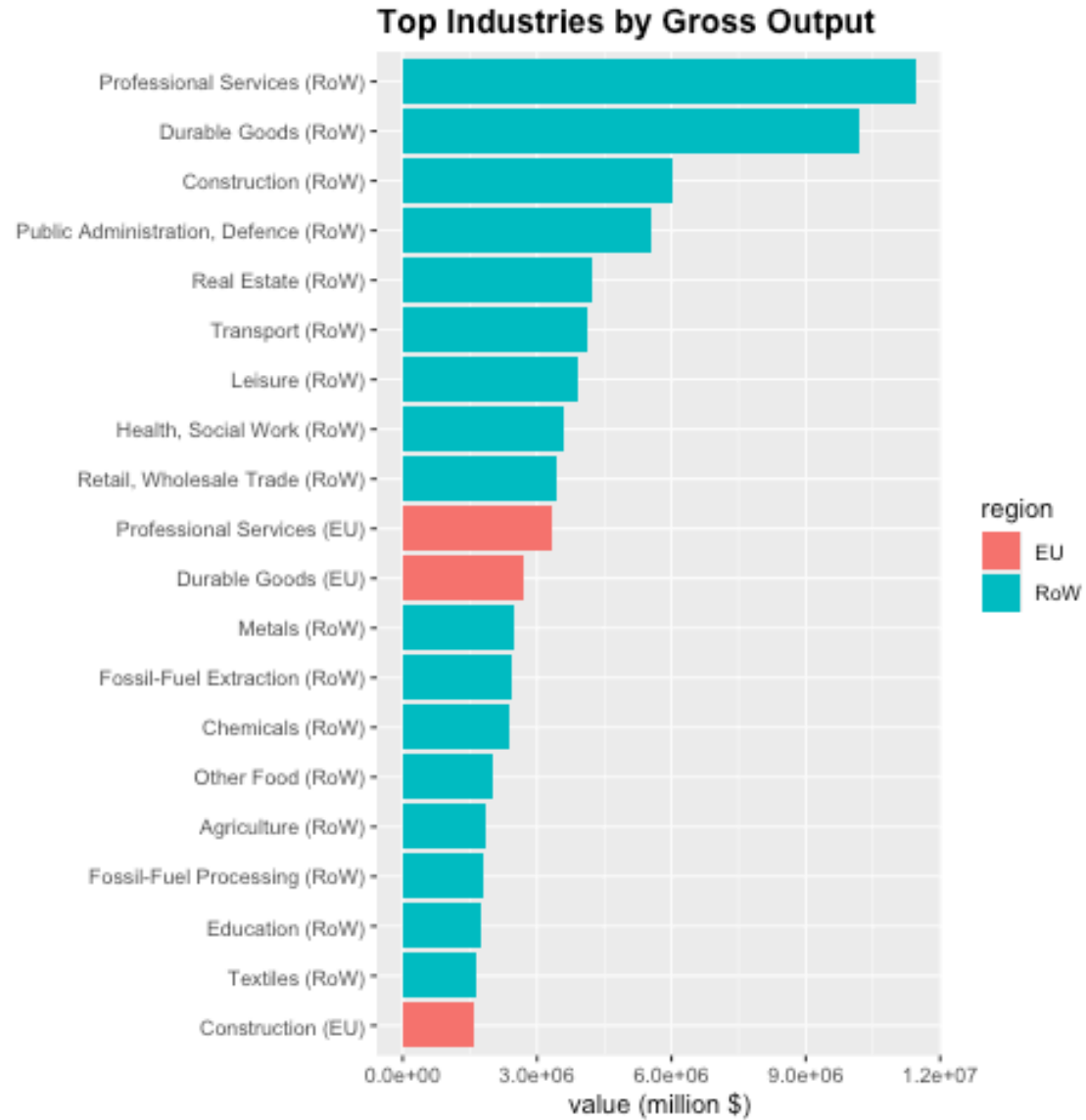
#	Industry
1	Agriculture
2	Animal Farming
3	Forestry, logging and related service activities (02)
4	Fishing, operating of fish hatcheries and fish farms; service activities incidental to fishing (05)
5	Extraction of Fossil-Fuels
6	Mining & Quarrying of raw materials
7	Processing of meat
8	Processing of other food & beverage
9	Manufacture of tobacco products (16)
10	Manufacture of textiles
11	Manufacture of wood and of products of wood and cork, except furniture; manufacture of articles of straw and plaiting materials (20)
12	Re-processing of secondary wood material into new wood material
13	Pulp
14	Re-processing of secondary paper into new pulp
15	Paper



Input-Output Network



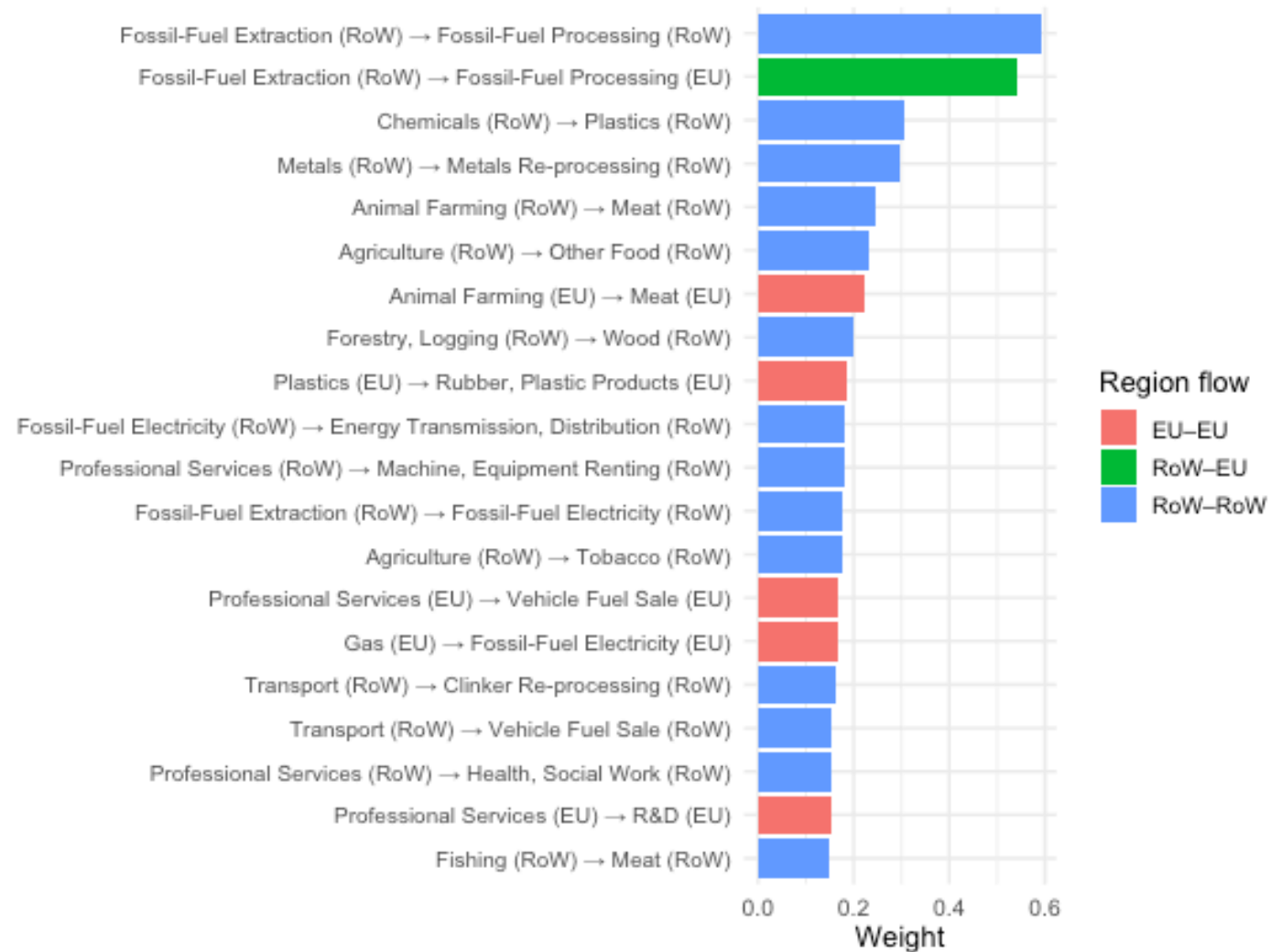
Top Nodes



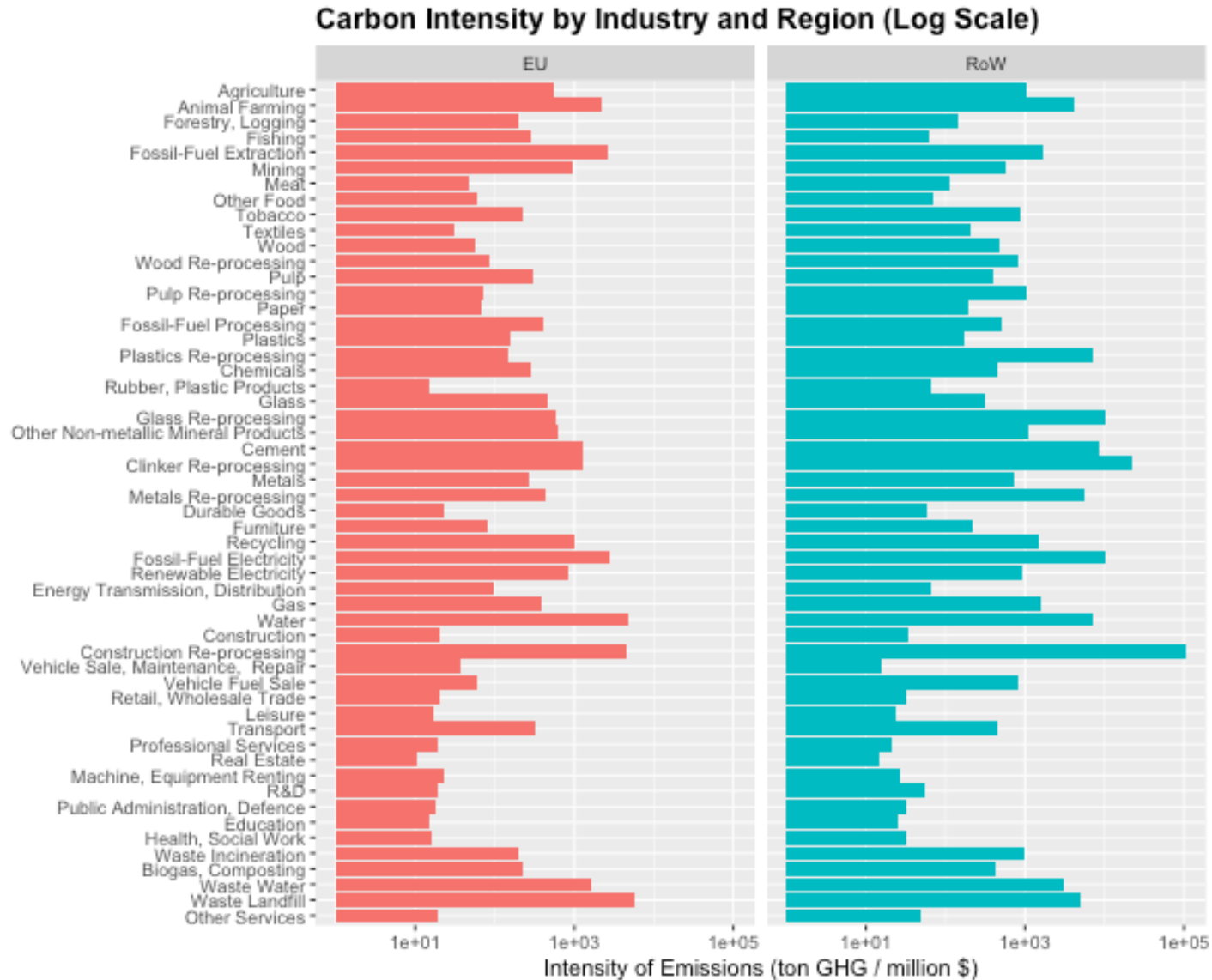
Top Links

Top 20 Links by IO Coefficients

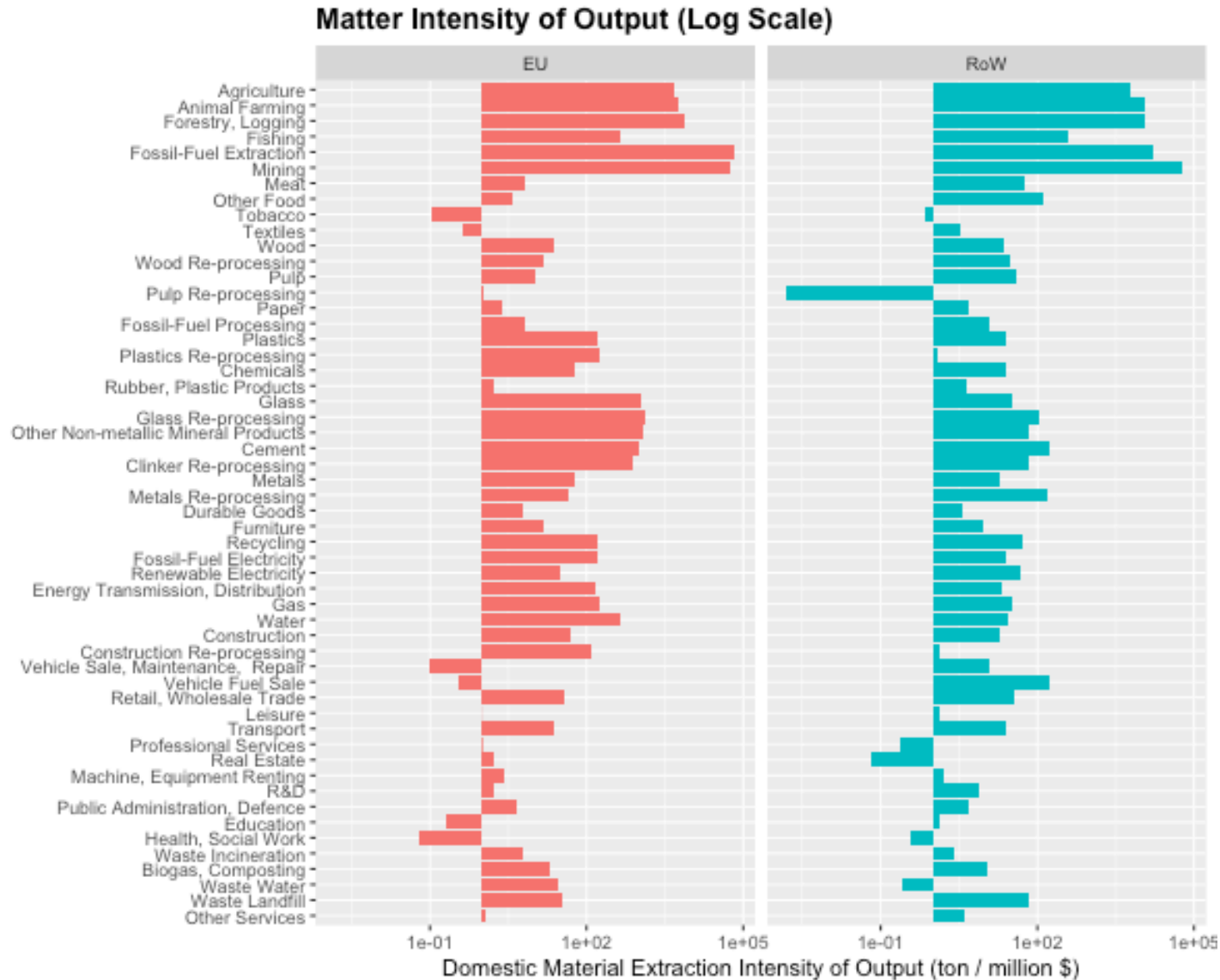
Colored by region-region flows



Carbon Intensity

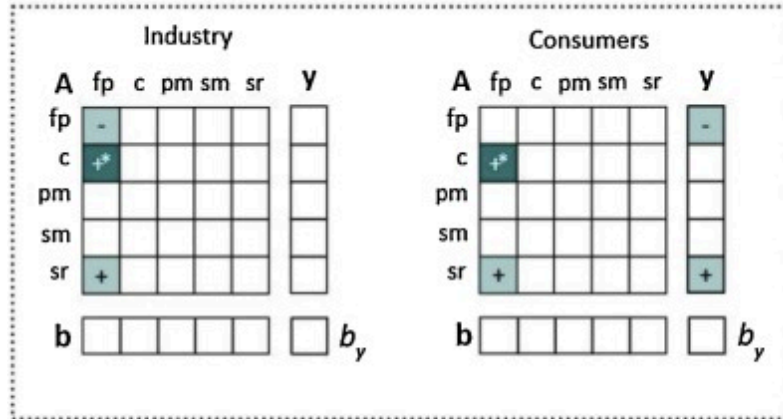


Matter Intensity



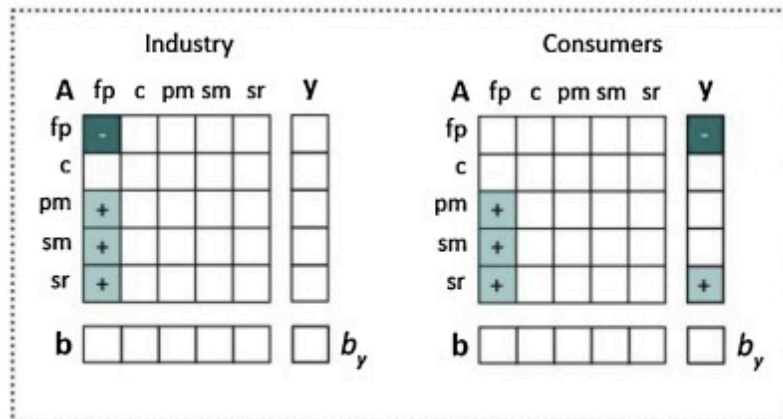
Modeling Circular Economy Interventions 1: Product Lifetime Extension

Reuse and remanufacturing (a)



A = Industry transactions
 y = Final demand
 b = intermediate extensions
 b_y = final demand extensions

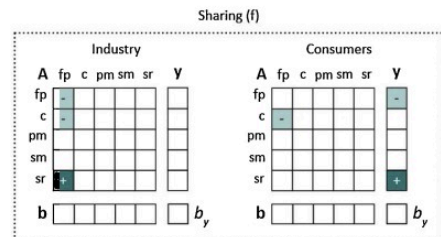
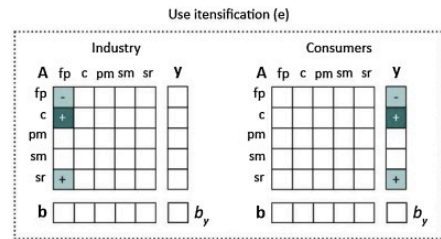
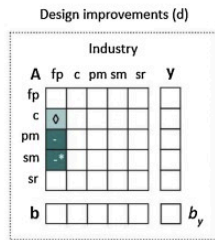
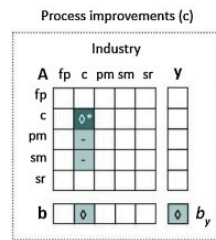
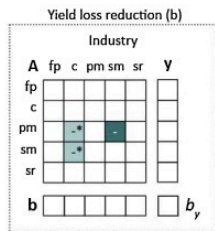
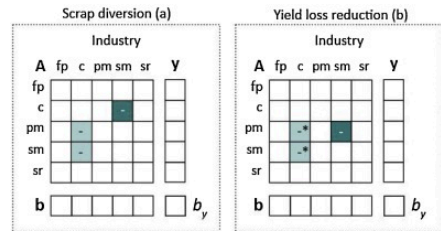
Delayed product replacement (b)



fp = final manufactured product
 c = component
 pm = primary material
 sm = secondary (recycled) material
 sr = services

- = reduction
 + = increase
 ◊ = unclear change
 * = substitution may apply
 ■ = primary change
 ■ = potential ancillary change

Modeling Circular Economy Interventions 2: Resource Efficiency



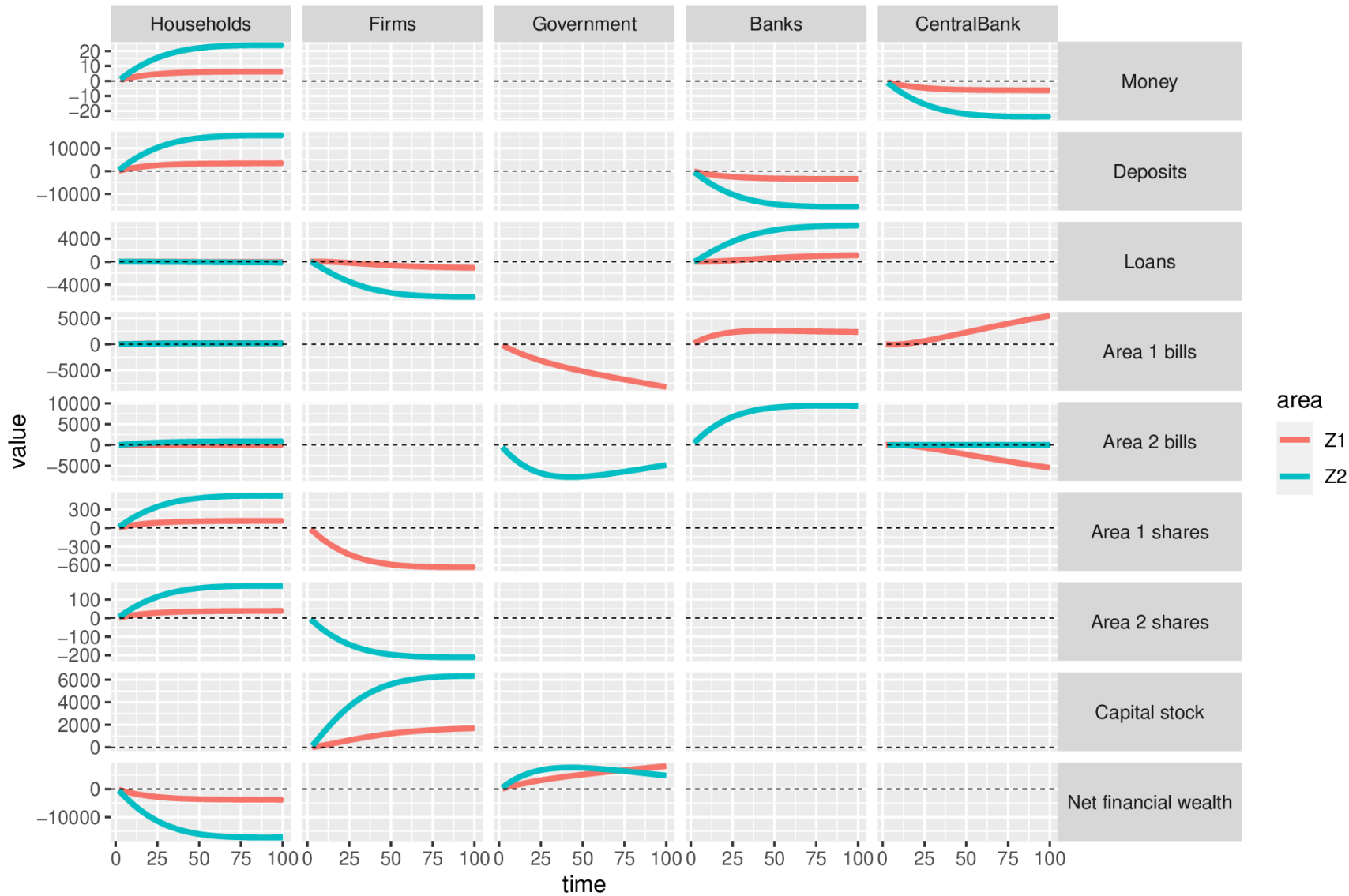
A = Industry transactions
 y = Final demand
 b = intermediate extensions
 b_y = final demand extensions

fp = final manufactured product
 c = component
 pm = primary material
 sm = secondary (recycled) material
 sr = services

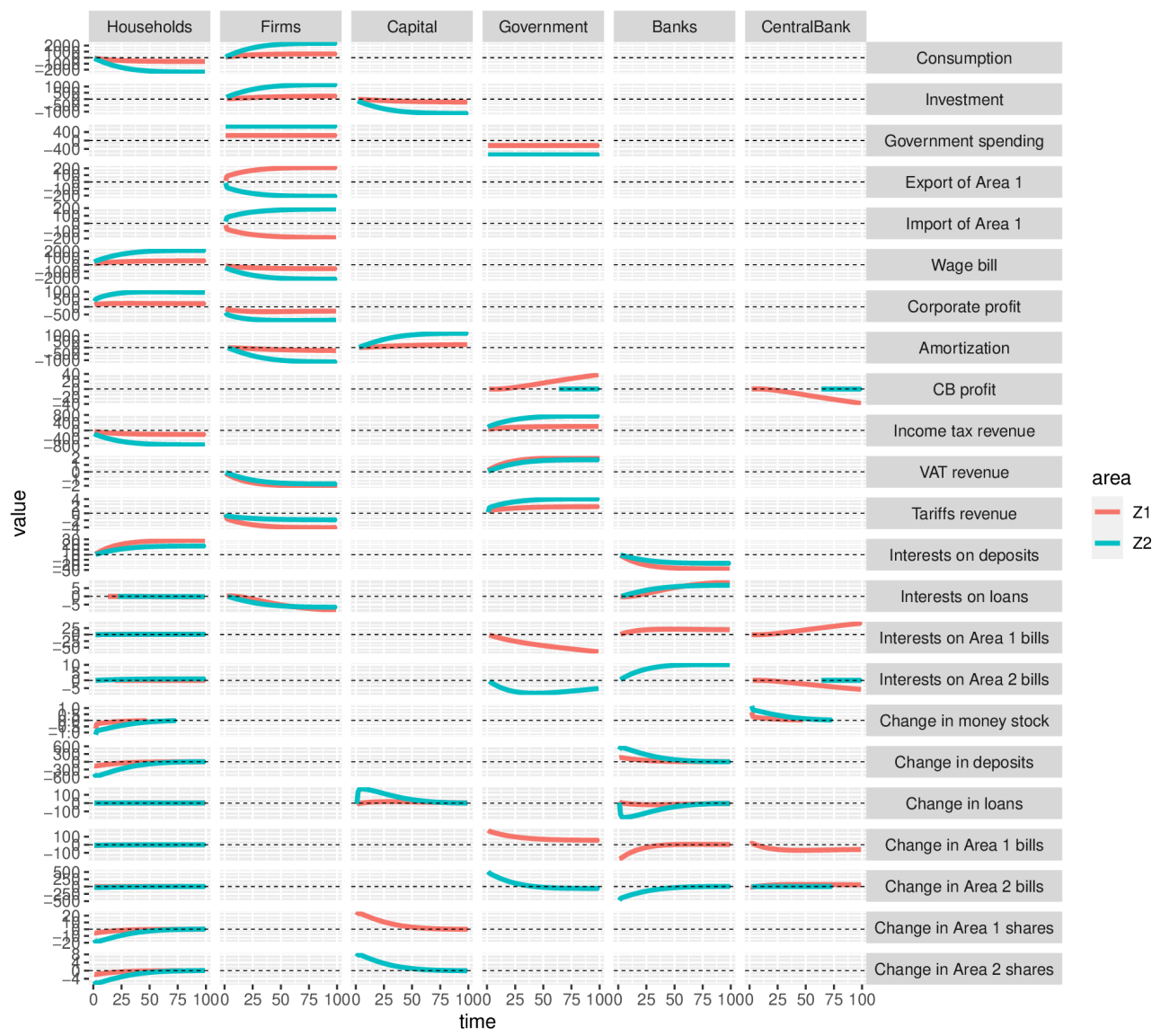
- = reduction
 + = increase
 $\diamond$ = unclear change
 * = substitution may apply
 ■ = primary change
 ■ = potential ancillary change

Results

Balance Sheet



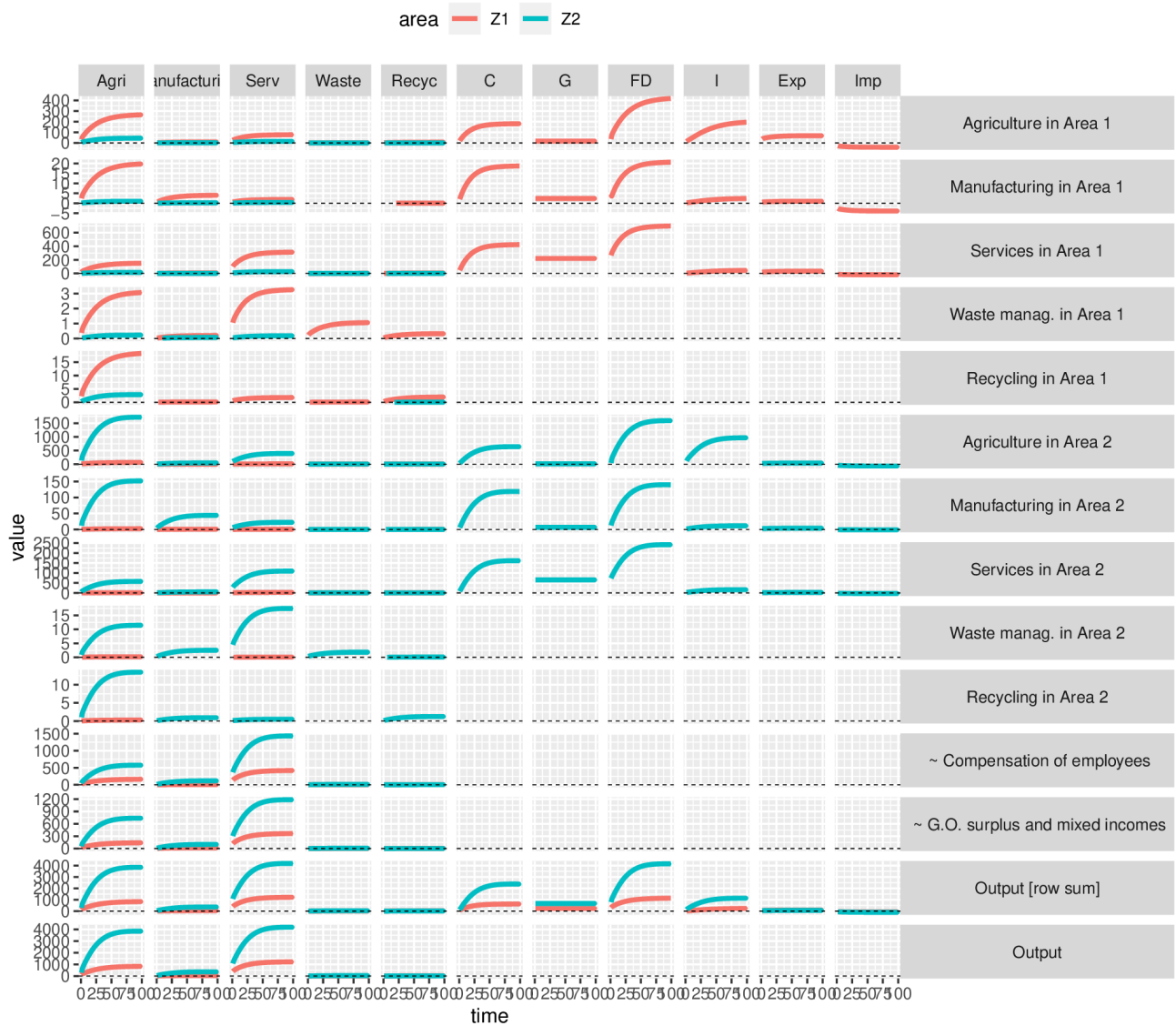
Transaction-Flow Matrix



TFM Formulas

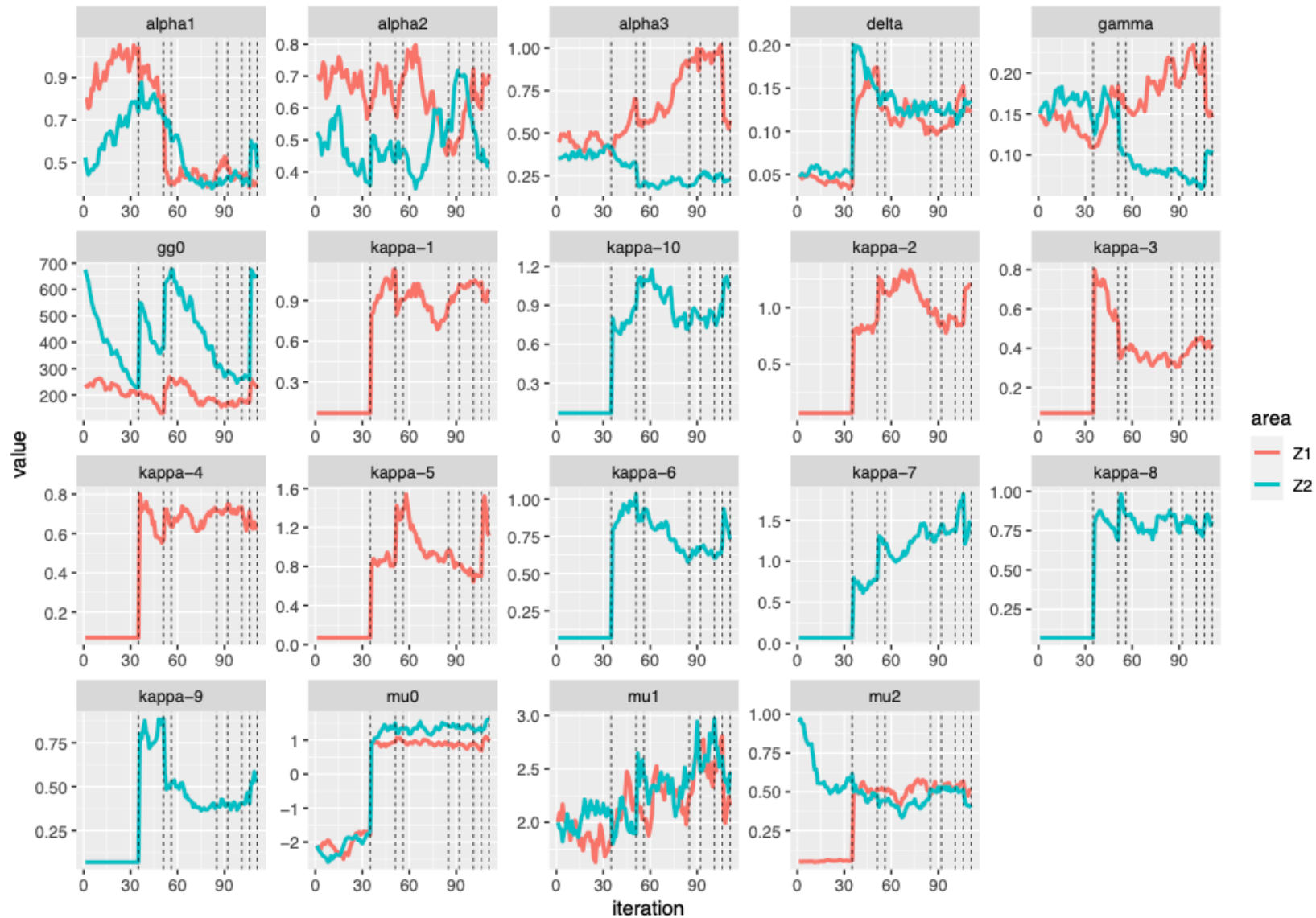
Search...																									
transaction	Firms_Current				Banks_Current				Central_Bank			Government													
Consumption	-	tfm	Consumption	Households																					
	+	tfm	ConsumptionofBankingservices	Households																					
	+	tfm	PublicServices	Households																					
Public Services	-	sna	P.13	S.13	R/P										+	sna	P.11	S.13	R/P						
	+	sna	P.3	S.13	U/A										-	tfm	PublicServices	FirmsCurrent							
Consumption of Banking services					-	sna	D.71	S.12	U/A						-	sna	D.71	S.121	U/A						
					+	sna	D.71	S.12	R/P						+	sna	D.71	S.121	R/P						
					-	sna	D.72	S.12	U/A						-	sna	D.72	S.121	U/A						
					+	sna	D.72	S.12	R/P						+	sna	D.72	S.121	R/P						
					-	tfm	ConsumptionofBankingservices	Households																	
Investment	-	tfm	Investment	FirmsCap																					
	-	tfm	Investment	BanksCurrent					-	sna	P.5b	S.12	U/A				-	sna	P.5b	U/A	-	sna	P.5b	S.13	U/A
	-	tfm	Investment	Government																					
	-	tfm	Investment	Households																					
Exports	-	tfm	Exports	RoW																					
Imports	-	tfm	Imports	RoW																					

Multi-Region Input-Output

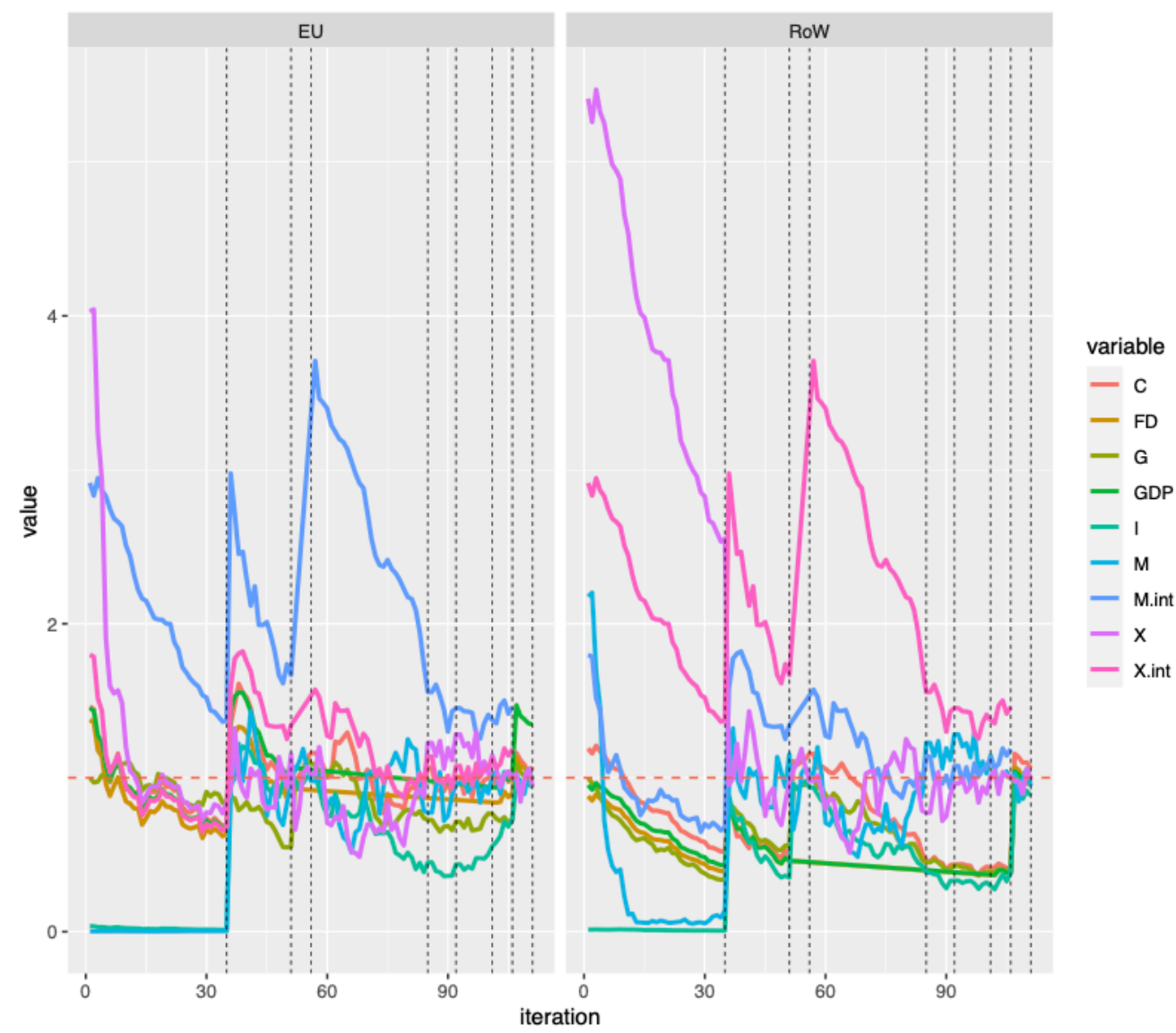


Calibration

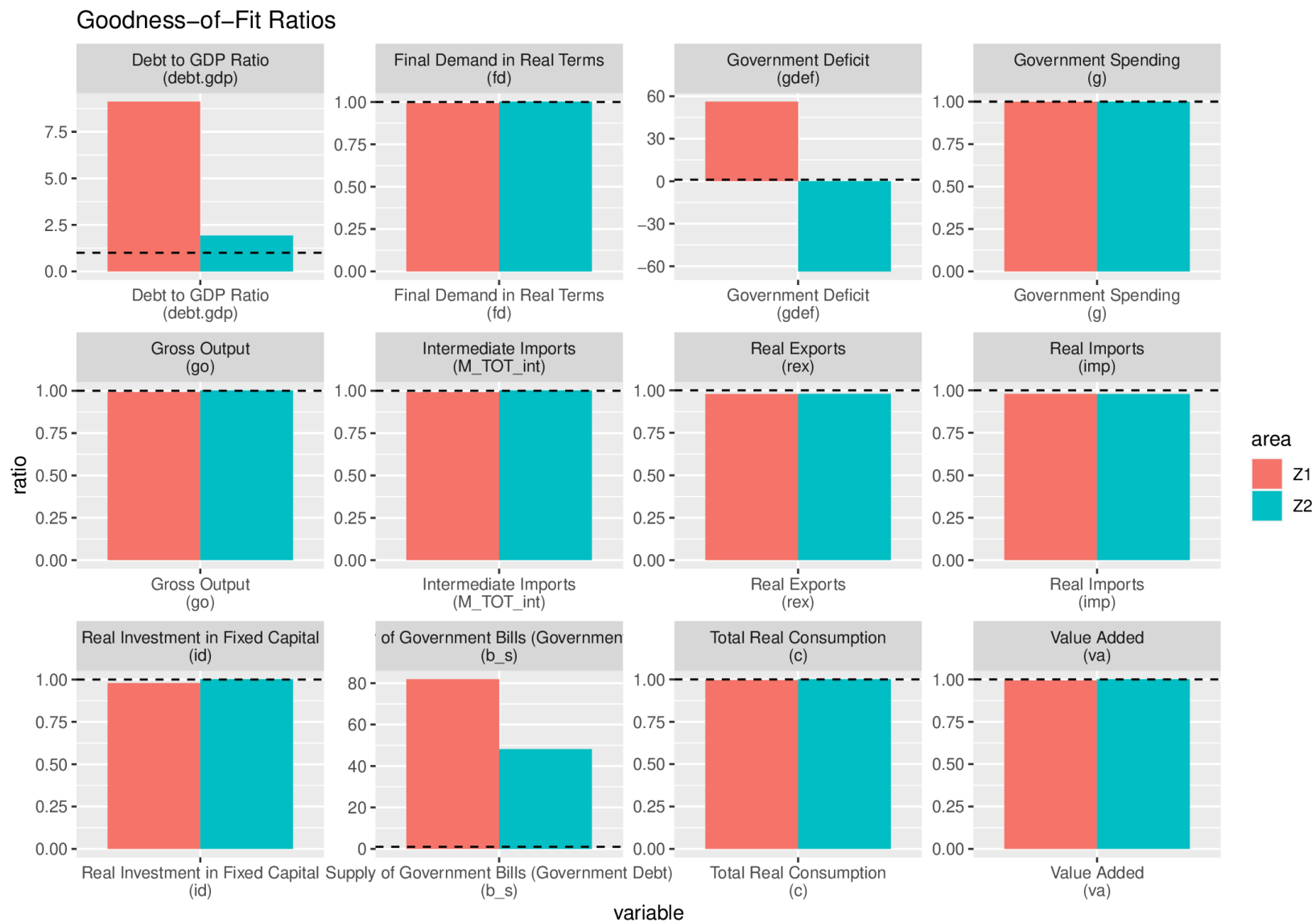
Instruments Evolution



Ratio Evolution



Calibration Result



Scenario Analysis

Shocks

Shock

Reduction in Consumption Level

Change in Consumption Composition towards Services

Product Life Time Extension

Higher Recycling Rate

Higher Propensity to Consume Green

Lower Extraction (or Conversion) Rate of Matter

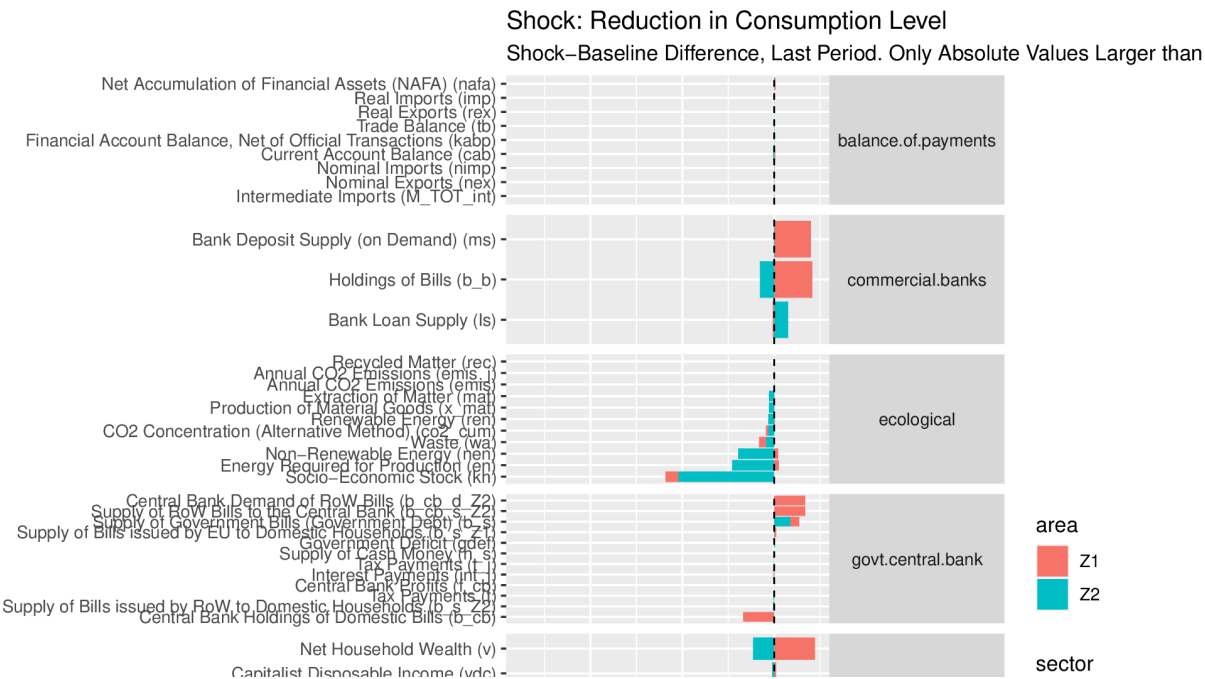
Lower Discarding Rate of Socio-Economic Stock

Higher Renewable Energy Share

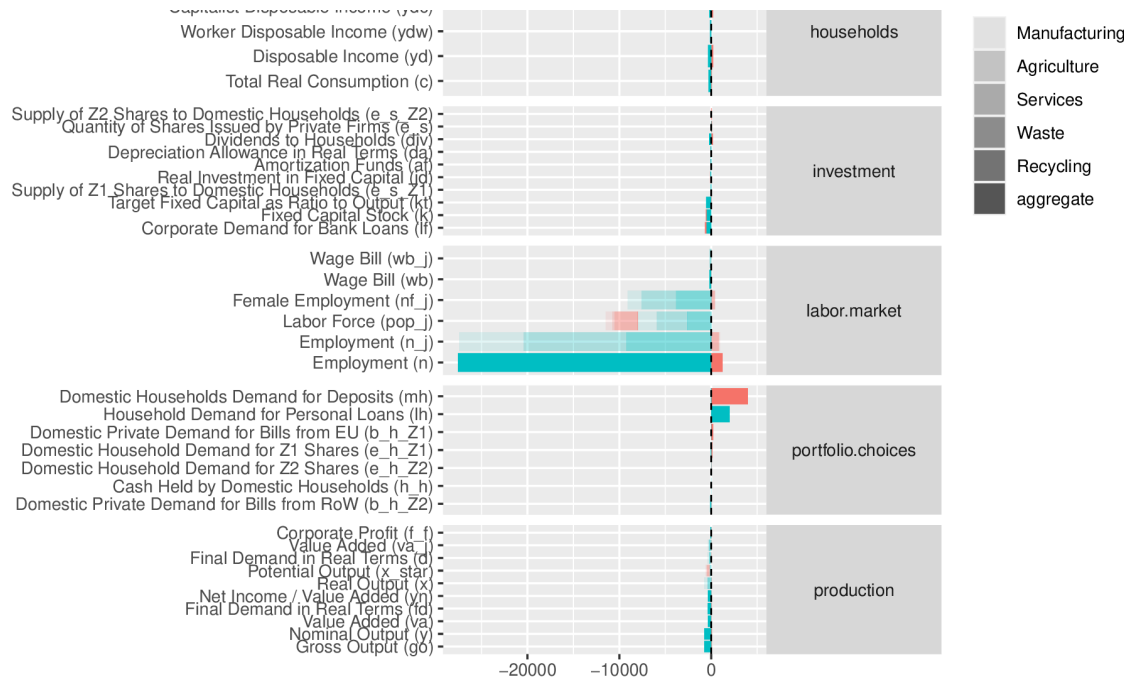
Higher Govt Spending towards Efficiency

Scenario: Reduction in Consumption Level

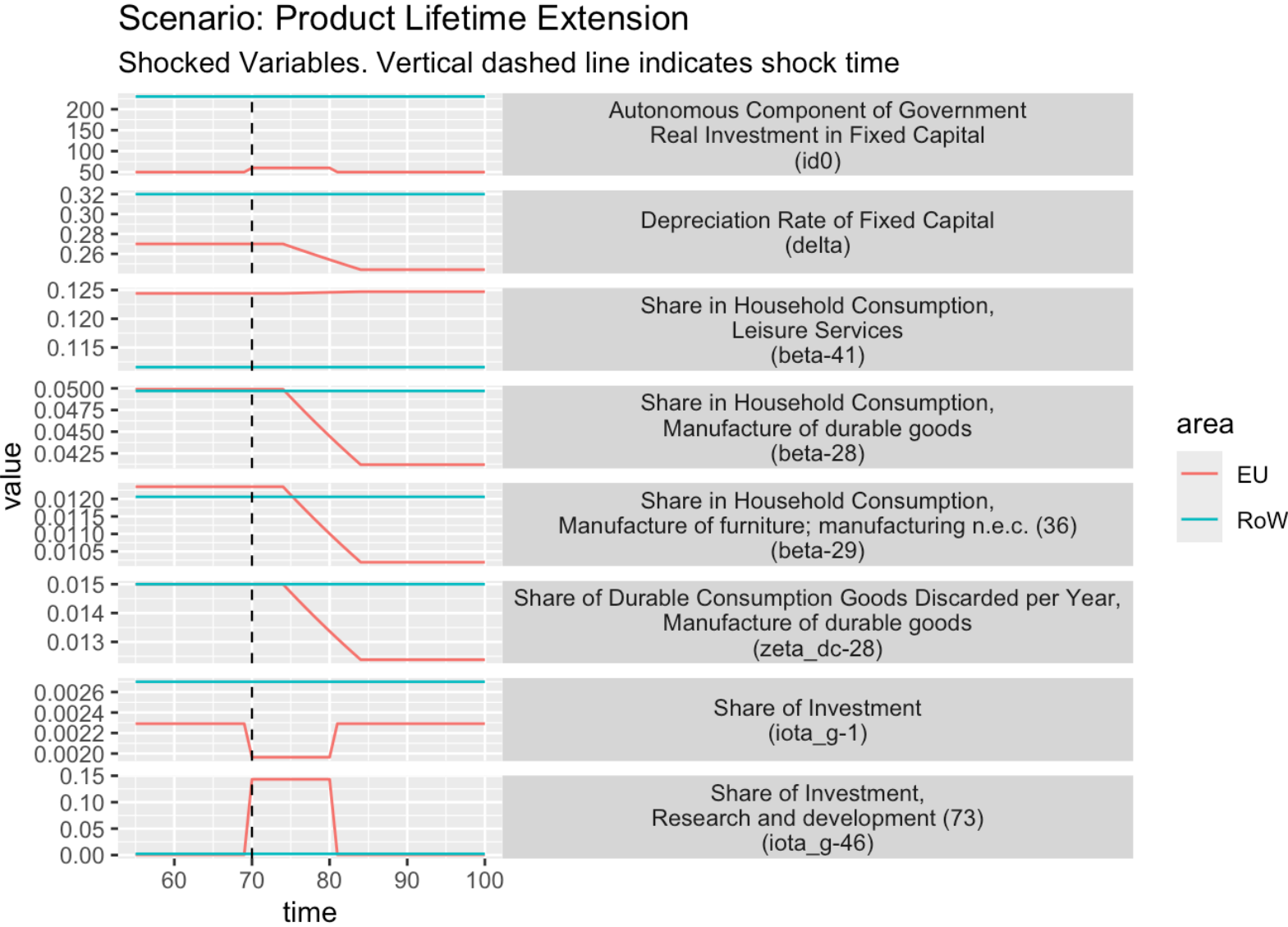
Top half



Bottom half

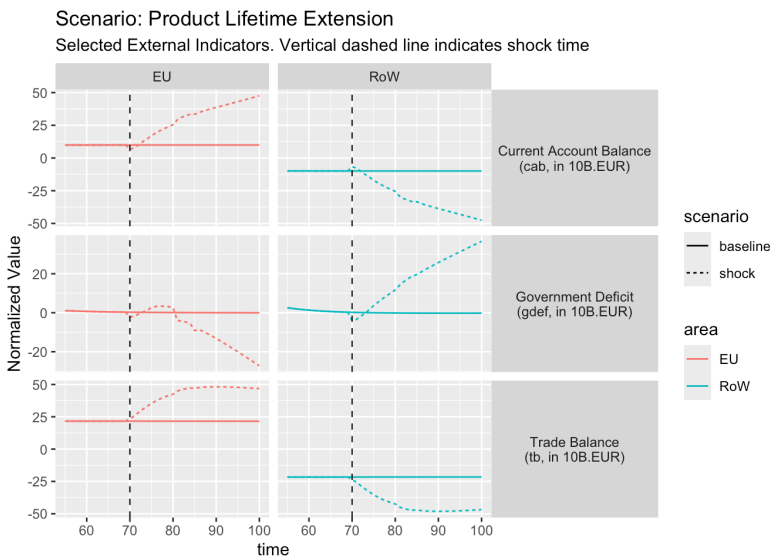


Scenario: Product Lifetime Extension

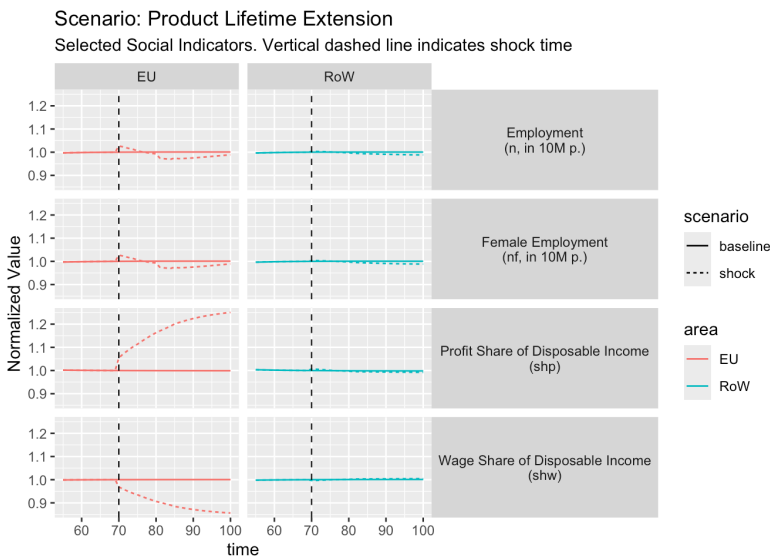


Impact: Product Lifetime Extension

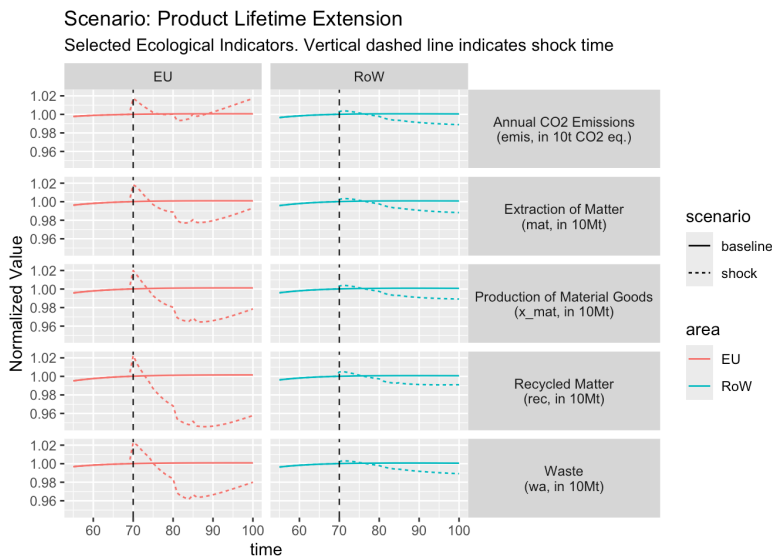
External Indicators



Social Indicators



Ecological Indicators

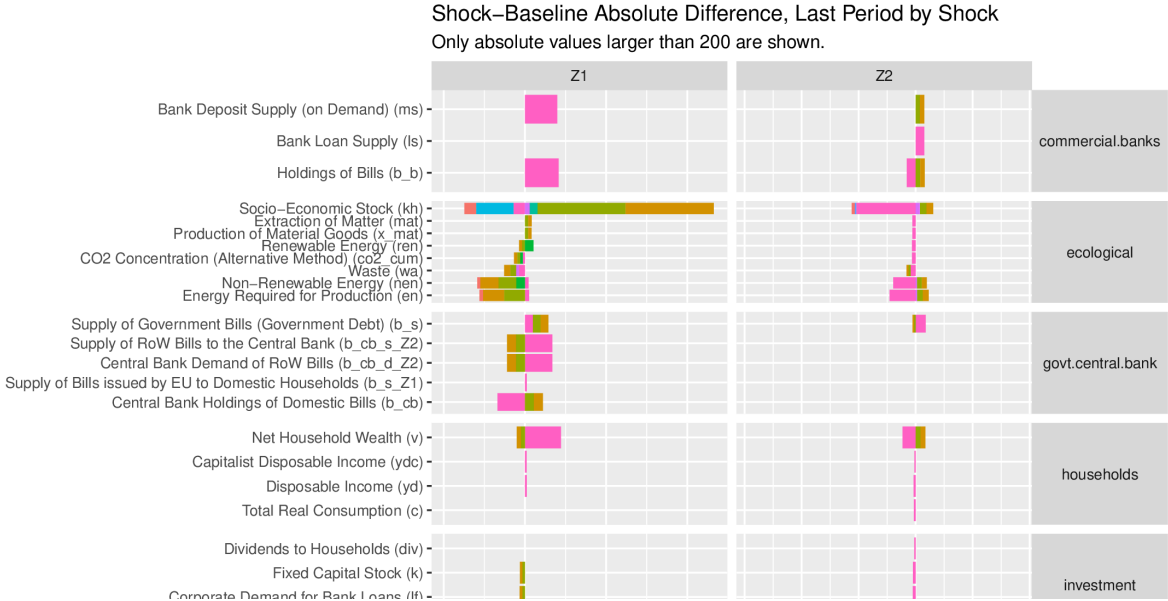


Government-led CE Transition



Scenario Summary: Change by Shock, Difference

Top half

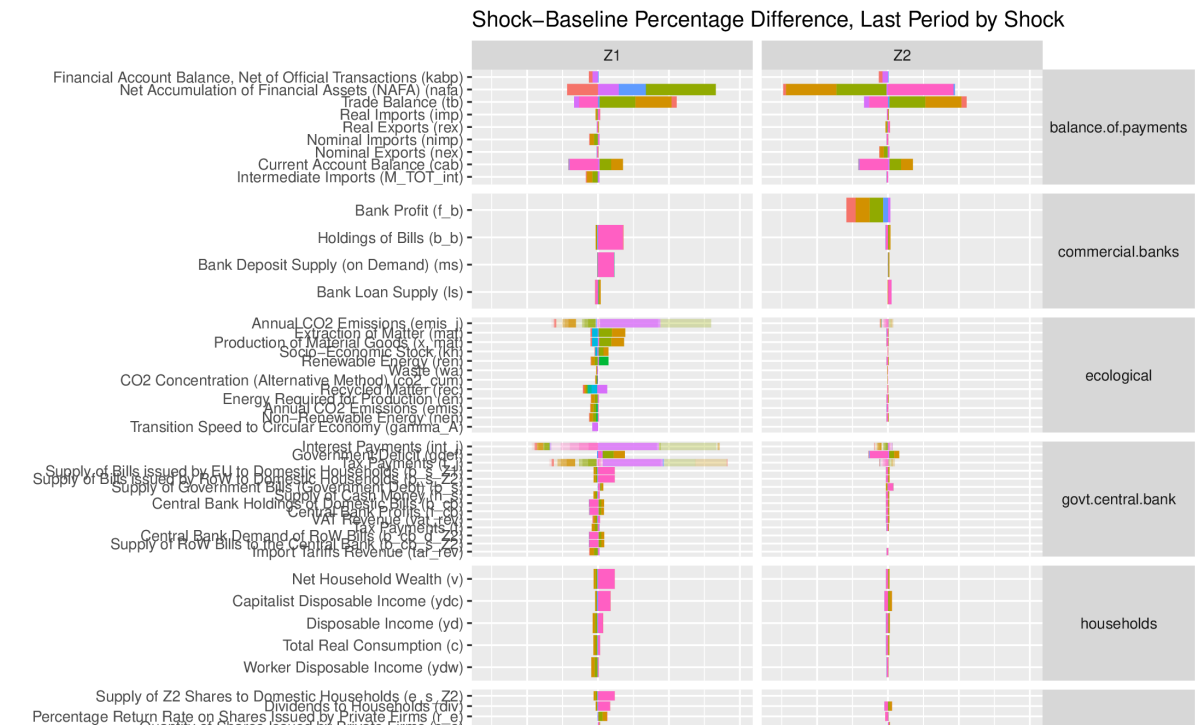


Bottom half

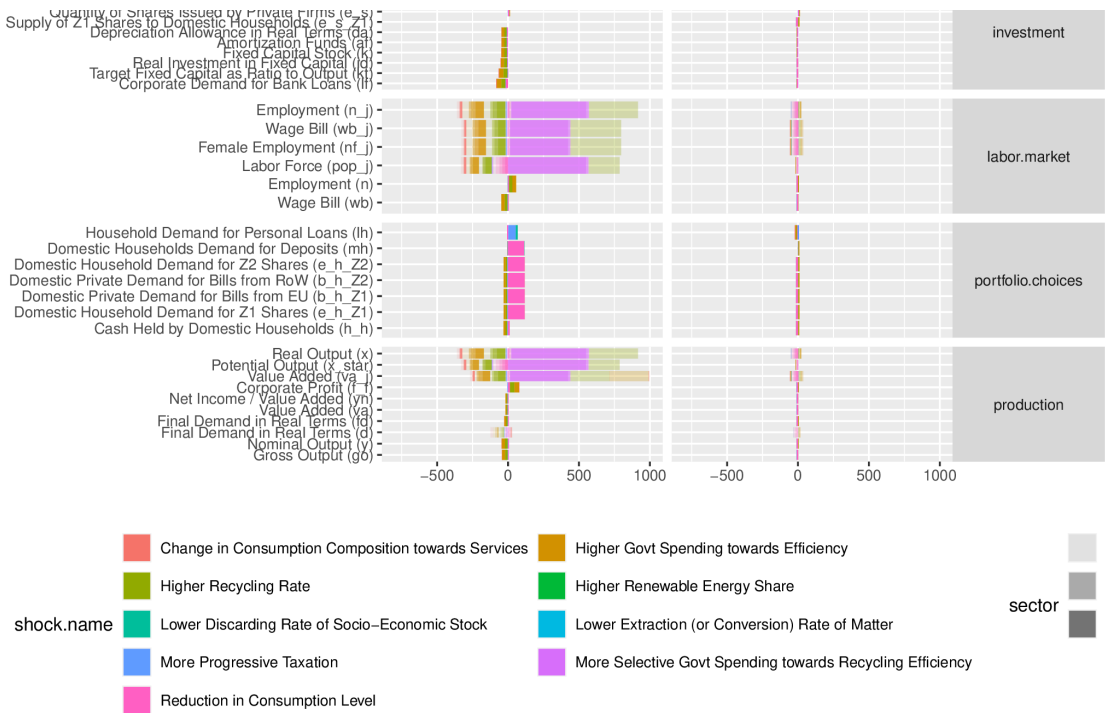


Scenario Summary: Change by Shock, Percent

Top half



Bottom half



Future Scenarios

Changes in Household Final Consumption Shares

- **Dietary transition:** reduced meat consumption (*Sector 7 → 8*)
- **Energy use transition:** (*Sector 31 → 32*)

Intermediate Production: **Shift from primary to secondary materials**

- Wood ($11 \rightarrow 12$), Pulp ($13 \rightarrow 14$), Plastics ($17 \rightarrow 18$)
- Metals ($26 \rightarrow 27$), Glass ($21 \rightarrow 22$), Cement ($24 \rightarrow 25$)
- Energy inputs ($31 \rightarrow 32$)
- Construction materials ($36 \rightarrow 27$)

Future Work

- Make explicit the military sector (military Keynesianism) and migration as a dynamic feedback
- **Green value chain upgrading** from the NZIPL green tech dataset
- NZIPL Net-Zero Value Chain Explorer

Thank you!

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References